# Software Requirements Specification

## for

## Student Attendance Management System (SAMS)

Version 1.1

Prepared by Team 1


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## Table of Contents

Revision History
1. Introduction
   1.1 Purpose
   1.2 Intended Audience
   1.3 Product Scope
   1.4 References
2. Overall Description
   2.1 Product Perspective
   2.2 Product Functions
   2.3 User Classes and Characteristics
   2.4 Operating Environment
   2.5 Design and Implementation Constraints
   2.6 Assumptions and Dependencies
3. External Interface Requirements
   3.1 User Interfaces
   3.2 Software Interfaces
   3.3 Communications Interfaces
4. Analysis Models
5. System Features
6. Other Nonfunctional Requirements
   6.1 Performance Requirements
   6.2 Safety Requirements
   6.3 Security Requirements
   6.4 Software Quality Attributes
   6.5 Business Rules
7. Other Requirements
Appendix A: Glossary
Appendix B: Field Layouts
Appendix C: Requirement Traceability Matrix

## Revision History

| Name | Date | Reason For Changes | Version |
|---|---|---|---|
| Team 1 | 03-10-2026 | Initial version | 1.0 |
| Team 1 | 29-10-2026 | Filled traceability matrix after implementation | 1.1 |

# 1. Introduction

## 1.1 Purpose

This document specifies the software requirements for the Student Attendance Management System (SAMS) version 1.0. SAMS is a web based application built using Python and Django for recording student attendance and generating attendance reports. This SRS covers the complete system.

## 1.2 Intended Audience

This document is for:

- Developers - to know what has to be built
- Testers - to write test cases from the requirements
- Faculty evaluator - to review the project
- Admin staff - who will be the actual users of the system

Section 2 gives the overall description, Section 3 the interface requirements, Section 4 the use case diagram, Section 5 the functional requirements and Section 6 the non functional requirements.

## 1.3 Product Scope

Currently attendance is taken on paper and later entered into excel sheets, which takes time and leads to mistakes in calculating percentages. Students also dont know thier attendance until the shortage list comes out at the end of the semester.

SAMS solves this by:

- letting faculty mark attendance online from a laptop or phone
- calculating attendance percentage automaticaly
- generating reports in PDF and Excel
- letting students check thier attendance anytime

- sending an email when attendance goes below 75%

### 1.4 References

- IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications
- Django Documentation - https://docs.djangoproject.com
- SE Mini Project Guidelines and Problem Statements, Section C

## 2. Overall Description

### 2.1 Product Perspective

SAMS is a new, standalone system that replaces the manual attendance register. It is not part of any existing system. It has three tiers:

| Tier | Technology |
|---|---|
| Front end | HTML, CSS, Bootstrap (Django templates) |
| Back end | Python, Django |
| Database | MySQL |

The system also connects to an external email (SMTP) server to send mails.

### 2.2 Product Functions

- Login and logout for Admin, Faculty and Student
- Manage students, faculty, subjects and classes
- Mark and update attendance
- View subject wise attendance
- Generate attendance reports and shortage list (PDF / Excel)
- Send shortage email to students below 75%
- Apply for leave and approve / reject leave

### 2.3 User Classes and Characteristics

| User Class | Description | Usage |
|---|---|---|
| Admin | College office staff / HOD. Manages students, faculty, subjects and classes. Full access. | Occasional |
| Faculty | Teachers. Mark attendance, generate reports, approve leave. | Daily |

| | | |
|---|---|---|
| | Most important user class. | |
| Student | Views own attendance and applies for leave. Mostly uses mobile. | Frequent |

### 2.4 Operating Environment

| Component | Requirement |
|---|---|
| Server OS | Windows 10+ or Ubuntu 20.04+ |
| Runtime | Python 3.10+, Django 4.2 |
| Database | MySQL 8.0 |
| Client | Any modern browser (Chrome, Firefox, Edge) on laptop or mobile |

### 2.5 Design and Implementation Constraints

- Must be developed in Python using Django (as per problem statement)
- MySQL must be used as the database
- Web application only, no mobile app
- Must be completed within the semester by a team of 4
- Code on GitHub, sprints tracked in JIRA

### 2.6 Assumptions and Dependencies

- College provides student and subject lists in Excel format
- A Gmail SMTP account is available for sending emails
- Faculty have a device with internet during class
- Depends on third party libraries: reportlab (PDF) and openpyxl (Excel)

## 3. External Interface Requirements

### 3.1 User Interfaces

- Web interface built with HTML, CSS and Bootstrap
- Role based dashboard shown after login
- Navigation bar with only the options allowed for that role
- Forms show error messages for invalid input
- Attendance page shows the student list with a checkbox for each student

- Pages work on both laptop and mobile screens

### 3.2 Software Interfaces

| Software | Purpose |
|---|---|
| MySQL 8.0 | Stores all data, accessed through Django ORM |
| Gmail SMTP | Sending shortage and leave emails |
| reportlab | Generating PDF reports |
| openpyxl | Reading student Excel uploads and generating Excel reports |

### 3.3 Communications Interfaces

- HTTPS between browser and server
- SMTP on port 587 with TLS for emails

## 4. Analysis Models

Figure 1 shows the use case diagram with the three actors - Admin, Faculty and Student.

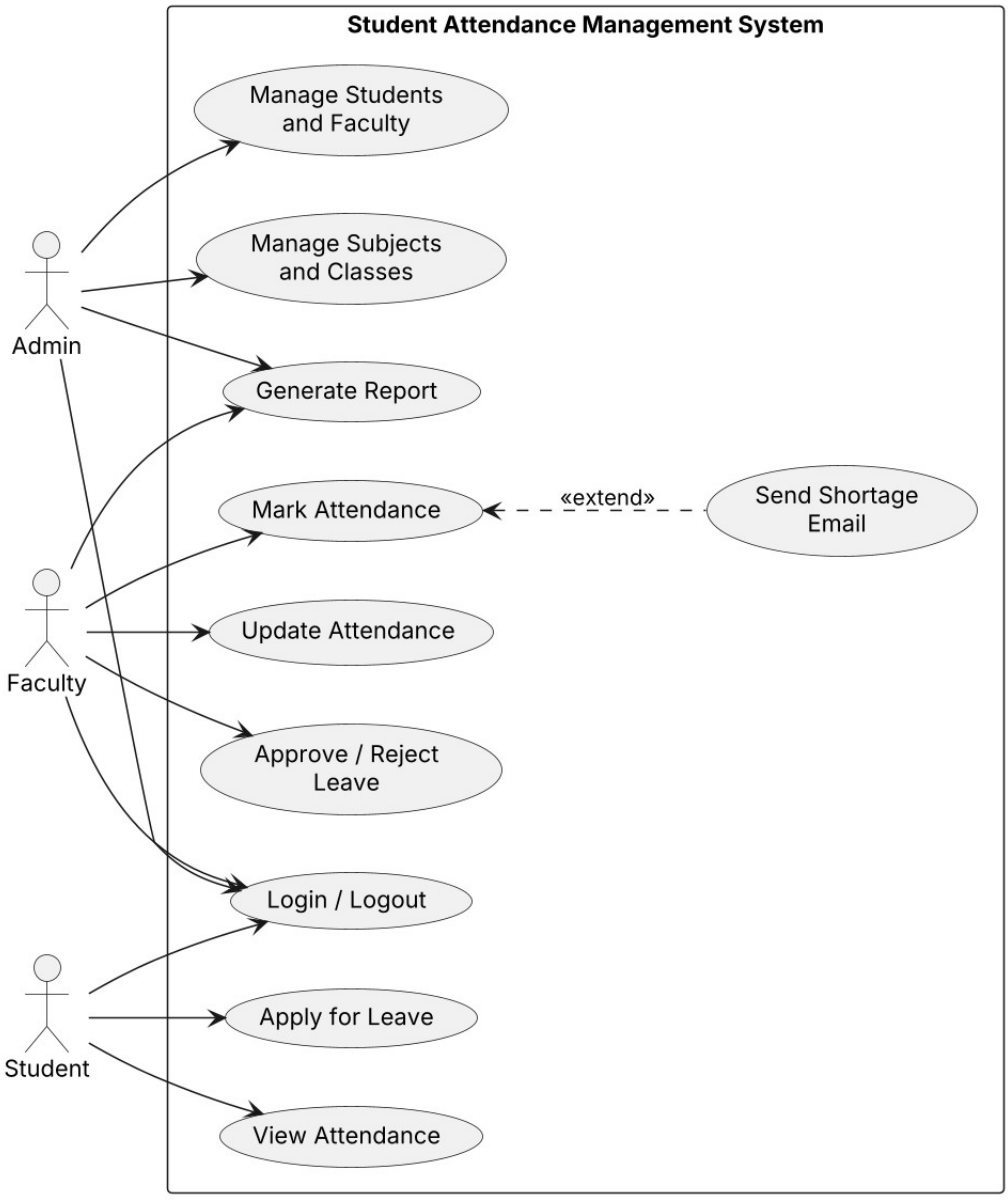



Figure 1: Use case diagram

# 5. System Features

## 5.1 User Login and Management

**Description and Priority:** Lets users login and lets admin manage user accounts. Priority - High.

**Stimulus / Response:**

| **User Action (Stimulus)** | **System Response** |
|---|---|
| User enters correct ID and password | Dashboard for that role is shown |
| User enters wrong password | "Invalid username or password" shown |

| | |
|---|---|
| | |
| Admin submits add student form | Student saved, success message shown |

**Functional Requirements:**

| ID | Requirement |
|---|---|
| REQ-1 | The system shall allow Admin, Faculty and Students to login using thier ID and password. |
| REQ-2 | The system shall show a different dashboard for each role after login. |
| REQ-3 | The Admin shall be able to add, edit and delete student and faculty accounts. |
| REQ-4 | The Admin shall be able to upload a list of students from an Excel file. |

## 5.2 Subject and Class Management

**Description and Priority:** Lets admin set up subjects and classes so attendance can be taken. Priority - High.

**Stimulus / Response:**

| User Action (Stimulus) | System Response |
|---|---|
| Admin adds subject and assigns faculty | Subject saved and shown on faculty dashboard |

**Functional Requirements:**

| ID | Requirement |
|---|---|
| REQ-5 | The Admin shall be able to add subjects with subject code, name and semester. |
| REQ-6 | The Admin shall be able to create classes (semester + section) and assign students to them. |
| REQ-7 | The Admin shall be able to assign a faculty to a subject for a class. |

### 5.3 Mark Attendance

**Description and Priority:** Lets faculty take and correct attendance for thier class. Priority - High.

**Stimulus / Response:**

| User Action (Stimulus) | System Response |
|---|---|
| Faculty selects subject, class, date and hour | Student list shown |
| Faculty marks absentees and submits | Attendance saved, confirmation shown |

**Functional Requirements:**

| ID | Requirement |
|---|---|
| REQ-8 | The Faculty shall be able to mark each student present or absent for a selected subject, class, date and hour. |
| REQ-9 | The system shall not allow attendance to be marked twice for the same subject, class, date and hour. |
| REQ-10 | The Faculty shall be able to update attendance within 3 days of marking it. |
| REQ-11 | The system shall recalculate attendance percentage of each student after every submission. |

### 5.4 View Attendance and Reports

**Description and Priority:** Lets students see attendance and faculty/admin generate reports. Priority - High.

**Stimulus / Response:**

| User Action (Stimulus) | System Response |
|---|---|
| Student opens My Attendance | Subject wise percentage shown |
| Faculty selects class, subject, dates and clicks Generate | Report created and downloaded |

**Functional Requirements:**

| ID | Requirement |
|---|---|
| REQ-12 | The Student shall be able to view thier attendance percentage for each subject. |
| REQ-13 | The Faculty and Admin shall be able to generate an attendance report for a class and subject between two dates. |
| REQ-14 | The report shall be downloadable in PDF and Excel format. |
| REQ-15 | The system shall generate a shortage list of students below 75% attendance. |

### 5.5 Shortage Email and Leave

**Description and Priority:** Warns students of low attendance and handles leave requests. Priority - Medium.

**Stimulus / Response:**

| User Action (Stimulus) | System Response |
|---|---|
| Student attendance drops below 75% | Email sent to student |
| Student submits leave form | Request sent to faculty |
| Faculty approves leave | Those days marked as Leave |

**Functional Requirements:**

| ID | Requirement |
|---|---|
| REQ-16 | The system shall send an email to the student when thier attendance in any subject falls below 75%. |
| REQ-17 | The Student shall be able to apply for leave by giving dates and reason. |
| REQ-18 | The Faculty shall be able to approve or reject a leave request. |
| REQ-19 | Approved leave days shall not be counted as absent. |

# 6. Other Nonfunctional Requirements

## 6.1 Performance Requirements

| ID | Requirement |
|---|---|
| NFR-1 | Every page shall load within 3 seconds with upto 100 users at the same time. |
| NFR-2 | Submitting attendance for a class of 70 students shall take less than 2 seconds. |
| NFR-3 | A full semester report for one class and subject shall be generated within 5 seconds. |

## 6.2 Safety Requirements

| ID | Requirement |
|---|---|
| NFR-4 | The database shall be backed up once every day. |
| NFR-5 | Attendance records shall never be permanantly deleted, only updated. |

## 6.3 Security Requirements

### Security Objectives:

| ID | Objective |
|---|---|
| SO-1 | Only authorised users can access the system and each user can only do what thier role allows. |
| SO-2 | Attendance data can only be changed by the faculty of that subject or the admin. |
| SO-3 | Personal details of students (email, phone) are kept confidential. |

### Security Requirements:

| ID | Requirement | Objective |
|---|---|---|
| NFR-6 | Passwords shall be stored hashed using Django PBKDF2, never in plain text. | SO-1 |
| NFR-7 | The system shall check the user role on every page and block pages not allowed for that role. | SO-1 |

| | | |
|---|---|---|
| NFR-8 | A student shall only be able to view thier own attendance. | SO-3 |
| NFR-9 | User session shall expire after 30 minutes of no activity. | SO-1 |
| NFR-10 | All forms shall be protected against CSRF and SQL injection. | SO-2 |
| NFR-11 | Every change to an attendance record shall be logged with faculty name, date and time. | SO-2 |

### 6.4 Software Quality Attributes

| **Attribute** | **Requirement** |
|---|---|
| Usability | A new faculty can mark attendance without any training. |
| Reliability | System available during college hours (8 AM - 5 PM) on working days. |
| Maintainability | Code split into seperate Django apps per module. |
| Portability | Runs on both Windows and Linux servers. |

### 6.5 Business Rules

| **ID** | **Rule** |
|---|---|
| BR-1 | Only Admin can add or delete students, faculty and subjects. |
| BR-2 | Only the faculty assigned to a subject can mark attendance for it. |
| BR-3 | Faculty can edit attendance only within 3 days; after that only Admin can. |
| BR-4 | Minimum 75% attendance per subject is needed for ESA eligibility (domain requirement). |
| BR-5 | Attendance % = (classes attended + approved leave) / classes conducted x 100 (domain requirement). |

## 7. Other Requirements

- Student data must be handled as per university data rules.
- Only English language is supported in this version.

## Appendix A: Glossary

| Term | Meaning |
|---|---|
| SAMS | Student Attendance Management System |
| SRN | Student Registration Number |
| ESA | End Semester Assessment |
| HOD | Head of Department |
| ORM | Object Relational Mapping |
| CSRF | Cross Site Request Forgery |
| SMTP | Simple Mail Transfer Protocol |
| Shortage | Attendance below 75% in a subject |

## Appendix B: Field Layouts

### Student Registration Fields

| Field | Length | Data Type | Description | Mandatory |
|---|---|---|---|---|
| SRN | 13 | Alphanumeric | Student registration number | Y |
| Name | 50 | Text | Full name | Y |
| Email | 50 | Alphanumeric | College email ID | Y |
| Phone | 10 | Numeric | Mobile number | N |
| Semester | 1 | Numeric | 1 to 8 | Y |
| Section | 1 | Text | Section | Y |
| Department | 5 | Text | e.g. CSE | Y |
| Password | 20 | Alphanumeric | Changed on first login | Y |

### Report Fields

| **Attendance Report** | **Shortage Report** |
|---|---|
| Subject Code and Name | Subject Code and Name |
| Class (Sem and Section) | Class (Sem and Section) |
| From Date / To Date | SRN |
| SRN | Student Name |
| Student Name | Classes Conducted |
| Classes Conducted | Classes Attended |
| Classes Attended | Attendance % |
| Leave Days | Email Sent (Y/N) |
| Attendance % | Faculty Name |
| Faculty Name | Generated On |
| Generated On | |

## Appendix C: Requirement Traceability Matrix

| **Sl. No** | **Req ID** | **Brief Description** | **Architecture Ref** | **Design Ref** | **Code File Ref** | **Test Case ID** | **System Test Case ID** |
|---|---|---|---|---|---|---|---|
| 1 | REQ-1 | allow Admin, Faculty and Students to login using thier ID and password | accounts | SAD 4.1 | accounts/ views.py | LoginTests | TC-01, 02 |
| 2 | REQ-2 | show a different dashboard for each role after login | accounts | SAD 3.9 | accounts/ decorators.py | DashboardTests | TC-03 |
| 3 | REQ-3 | Admin can add, edit and delete student and faculty accounts | accounts | SAD 3.4 | accounts/ admin.py | - | TC-05 |
| 4 | REQ-4 | Admin can upload a list of students from an Excel file | accounts | SAD 4.3 | accounts/ importer.py | StudentExcelUploadTests | TC-04 |
| 5 | REQ-5 | Admin can add subjects with subject code, name and semester | academics | SAD 3.6 | academics/ admin.py | AcademicsConstraintTests (manual: add via | TC-05 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | | admin) | |
| 6 | REQ-6 | Admin can create classes (semester + section) and assign students to them | academics | SAD 3.6 | academics/ admin.py | AcademicsConstraintTests | TC-05 |
| 7 | REQ-7 | Admin can assign a faculty to a subject for a class | academics | SAD 3.6 | academics/ models.py | AcademicsConstraintTests | TC-05 |
| 8 | REQ-8 | Faculty can mark each student present or absent for a selected subject, class, date and hour | attendance | SAD Fig 2 | attendance/ views.py | MarkAttendanceTests | TC-06 |
| 9 | REQ-9 | not allow attendance to be marked twice for the same subject, class, date and hour | attendance | SAD Fig 2 | attendance/ views.py | MarkAttendanceTests | TC-07 |
| 10 | REQ-10 | Faculty can update attendance within 3 days of marking it | attendance | SAD 4.3 | attendance/ views.py | EditAttendanceTests | TC-08 |
| 11 | REQ-11 | recalculate attendance percentage of each student after every submission | attendance | SAD 4.1 | attendance/ utils.py | AttendancePercentTests | TC-09 |
| 12 | REQ-12 | Student can view thier attendance percentage for each subject | reports | SAD 4.3 | reports/views.py | ApproveLeaveTests (% shown on My attendance) | TC-10 |
| 13 | REQ-13 | Faculty/Admin can generate an attendance report for a class and subject between two dates | reports | SAD 4.3 | reports/ services.py | ClassReportTests | TC-11 |
| 14 | REQ-14 | Report downloadable in PDF and Excel format | reports | SAD 4.3 | - | - | TC-11 |
| 15 | REQ-15 | generate a shortage list of students below 75% attendance | reports | SAD 4.3 | reports/views.py | ShortageListTests | TC-12 |
| 16 | REQ-16 | send an email to the student when thier attendance in any subject falls below 75% | leaves | SAD Fig 2 | leaves/utils.py | ShortageEmailTests | TC-13 |
| 17 | REQ-17 | Student can apply for leave | leaves | SAD Fig | leaves/views.py | ApplyLeaveTest | TC-14 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | by giving dates and reason | | 3 | | s | |
| 18 | REQ-18 | Faculty can approve or reject a leave request | leaves | SAD Fig 3 | leaves/views.py | ApproveLeaveTests | TC-14 |
| 19 | REQ-19 | Approved leave days shall not be counted as absent | leaves | SAD Fig 3 | leaves/views.py | ApproveLeaveTests | TC-14 |